

Multi-state models: Beware the Markov assumption!

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Competing risks and multi-state models course
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Two parts, both relatively new work

Landmark Aalen-Johansen estimator (LMAJ)

- ▶ Published in Stat Methods Med Res (2018)
- ▶ Joint work with Cristian Spitoni

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Landmark Aalen-Johansen estimator (LMAJ)

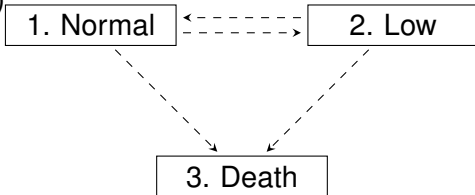
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Follow-up work

- ▶ Tests for the Markov assumption, Biostatistics (2020)
 - ▶ Joint work with Andrew Titman
- ▶ Hybrid LMAJ
 - ▶ More efficient and robust estimation of transition probabilities
 - ▶ Very recent work with the Oslo group, Lifetime Data Anal (2021)

CSL1 trial

- ▶ Placebo-controlled randomized clinical trial
- ▶ To evaluate the efficacy of prednisone in patients with histologically verified liver cirrhosis
- ▶ Prothrombin index, indicator of liver functioning
 - ▶ Dichotomized into normal and low (less than 70% of normal values)
 - ▶ Changes between normal and low assumed to take place at measurement
- ▶ Studied in the context of a multi-state model (Andersen et al. 1993)



Notation

- ▶ State space $\mathcal{K} = \{1, \dots, K\}$, finite
- ▶ Random process $\{X(t); t \geq 0\}$ taking values in $\mathcal{K}$
- ▶ **History:** The sigma-algebra generated by the observed trajectories $\mathcal{H}_t$

Important concepts

► Transition intensities:

$$\lambda_{jk}(t | \mathcal{H}_{t-}) = \lim_{\Delta t \downarrow 0} P(X(t + \Delta t) = k | X(t) = j, \mathcal{H}_{t-}) / \Delta t.$$

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$$P_{\ell m}(s, t | \mathcal{H}_{s-}) = P(X(t) = m | X(s) = \ell, \mathcal{H}_{s-}).$$

- This is the probability that the process is in state m at time t , given that it was in state ℓ at time s and conditionally on the past trajectory until time s
- Note: the transition intensities only concern **direct** transitions, while transition probabilities might go over several transitions

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- **State occupation probabilities:**

$$P_m(t) = P(X(t) = m)$$

Markov property

- ▶ The multi-state process is a **Markov process** if it fulfills the **Markov property**:

$$P(X(t) = m | X(s) = \ell, \mathcal{H}_{s-}) = P(X(t) = m | X(s) = \ell)$$

- ▶ Future behavior of the process depends on the history only through the present

If the Markov property holds

- ▶ The transition intensities/hazards and the transition probabilities do not depend on the further past anymore, so

$$\lambda_{jk}(t) = \lim_{\Delta t \downarrow 0} P(X(t + \Delta t) = k | X(t) = j) / \Delta t$$

$$P_{\ell m}(s, t) = P(X(t) = m | X(s) = \ell)$$

Relation intensities and probabilities

- ▶ The Markov assumption is very convenient, mathematically, because:
- ▶ If the multi-state model is Markov, then

$$\mathbf{P}(s, t) = \prod_{s < u \leq t} (\mathbf{I} + d\mathbf{\Lambda}(u))$$

- ▶ $\mathbf{P}(s, t)$, $K \times K$ matrix with (ℓ, m) th element $P_{\ell m}(s, t)$
- ▶ $\mathbf{\Lambda}(t)$ also $K \times K$ matrix
 - ▶ (j, k) th element $\Lambda_{jk}(t) = \int_0^t \lambda_{jk}(s) ds$
 - ▶ (j, j) th diagonal element $\Lambda_{jj}(t) = -\sum_{k \neq j} \Lambda_{jk}(t)$

State occupation probabilities

- ▶ Vector $\mathbf{P}(t)$ of **state occupation probabilities**
 - ▶ m th element $P_m(t) = P(X(t) = m)$
 - ▶ Can be retrieved through $\mathbf{P}(t) = \pi(0)\mathbf{P}(0, t)$
 - ▶ $\pi(0)$ a $1 \times K$ vector with k th element $\pi_k(0) = P(X(0) = k)$
- ▶ Together:

$$\mathbf{P}(t) = \pi(0) \prod_{0 < u \leq t} (\mathbf{I} + d\Lambda(u))$$

- ▶ Product integral relation similar to

$$S(t) = \prod_{0 < u \leq t} (1 - d\Lambda(u))$$

Estimation of transition intensities

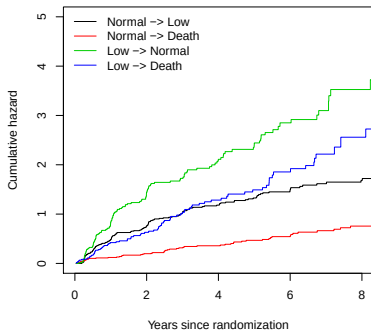
- ▶ Observe right censored versions of the multi-state process
- ▶ Non-parametric estimate of transition intensity $j \rightarrow k$:

$$\hat{\Lambda}_{jk}(t) = \int_0^t Y_j^{-1}(u) dN_{jk}(u)$$

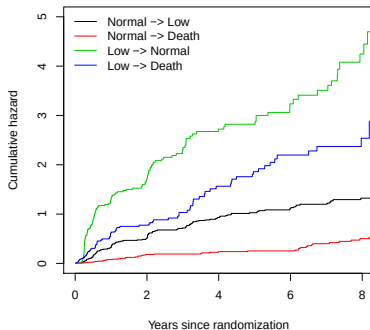
- ▶ Extension of the Nelson-Aalen estimator in standard survival analysis

Transition intensities

Placebo



Prednisone



The Aalen-Johansen estimator

- ▶ Gather estimated transition intensities in matrices $\hat{\Lambda}$
- ▶ Aalen-Johansen estimator uses relationship between transition intensities and probabilities
 - ▶ Replacing Λ by estimate $\hat{\Lambda}$

$$\hat{\mathbf{P}}(s, t) = \prod_{(s, t]} \left(\mathbf{I} + \Delta \hat{\Lambda}(u) \right)$$

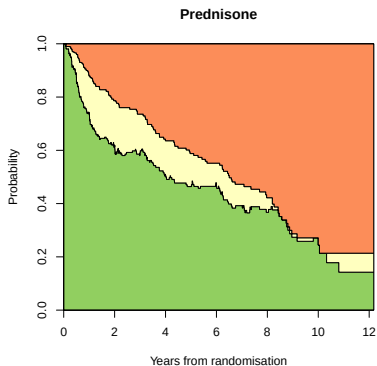
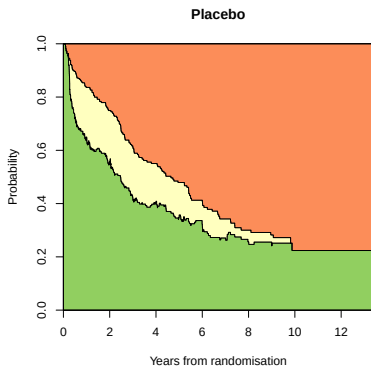
- ▶ Estimate of state occupation probabilities

$$\hat{\mathbf{P}}(t) = \hat{\pi}(0) \prod_{0 < u \leq t} \left(\mathbf{I} + \Delta \hat{\Lambda}(u) \right)$$

with $\hat{\pi}_k(0) = n^{-1} \sum_{i=1}^n 1\{X_i(0) = k\}$

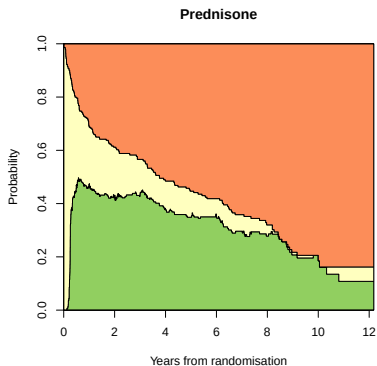
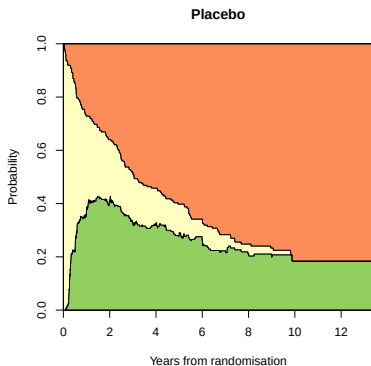
Transition probabilities

Given normal prothrombin level at baseline



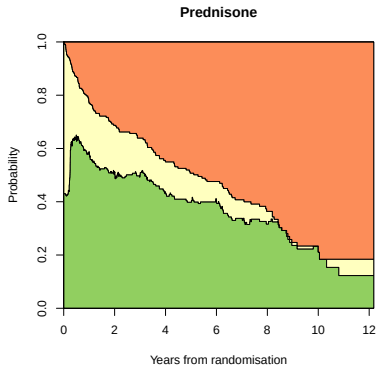
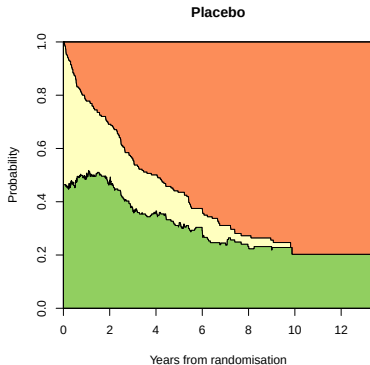
Transition probabilities

Given low prothrombin level at baseline



State occupation probabilities

- At baseline 46.4% of placebo and 43.0% of prednisone in normal state



Back to the Markov property

- ▶ The Aalen-Johansen is valid for arbitrary multi-state models
- ▶ As long as they are Markovian
- ▶ The Markov property facilitates elegant mathematical theory connecting transition intensities and transition probabilities
- ▶ It is therefore often assumed (taken for granted)
- ▶ But not at all obvious
- ▶ Violations
 - ▶ Duration dependence (semi-Markov)
 - ▶ “State visit” dependence
- ▶ Unfortunately, in our application, it turns out that normal low hazard increases with number of previous “lows” →

Transition probabilities for non-Markovian multi-state models

- ▶ When the multi-state model is not Markov, $P_{\ell m}(s, t | \mathcal{H}_{s-})$ will depend on the past before time s
 - ▶ Markov renewal process: $P_{\ell m}(s, t | \mathcal{H}_{s-})$ will crucially depend on the time at which state ℓ was entered before s
- ▶ When nature of violation of Markov is known, it is of course better to take that into account
- ▶ Transition probability $P_{\ell m}(s, t)$ still of interest when
 1. Extra relevant information in $\mathcal{H}_{s-}$ is not available
 2. When it is unknown how $\lambda_{jk}(t; \mathcal{H}_{t-})$ depends on the history
 - ▶ In practice, often uncertain whether or not the multi-state model is Markov
 - ▶ Useful to have estimator that is robust against possible non-Markovianity

Recent work

- ▶ After earlier work by Meira-Machado et al. (2006) and Allignol et al. (2014)
- ▶ Two recent papers, both Biometrics in 2015
- ▶ Both based on subsampling approach: selection is made of the data consisting of subjects occupying a given state at a particular time
- ▶ de Uña Álvarez and Meira-Machado (2015)
 - ▶ Irreversible illness-death model, extension to general forward-going multi-state models
 - ▶ Works with (differences between) Kaplan-Meiers
- ▶ Titman (2015)
 - ▶ Valid for any multi-state model
 - ▶ Based on derived competing risks processes on a "per ℓm basis"
 - ▶ As a result, the estimators of $\hat{P}_{lm}(s, t)$ are not guaranteed to add up to 1

Datta & Satten (2001)

- ▶ Datta & Satten (2001) showed that

$$\hat{\mathbf{P}}(t) = \hat{\pi}(0) \prod_{0 < u \leq t} (\mathbf{I} + \Delta \hat{\mathbf{\Lambda}}(u))$$

provides a consistent estimator of the state occupation probabilities $P(X(t) = m)$, **also if the Markov assumption is violated**

- ▶ Note: the estimators $\hat{\mathbf{\Lambda}}(u)$ are exactly as used before in the Markov model, but now have an interpretation as *partly conditional rates*
- ▶ Can we exploit this result to robustly estimate (partly conditional) transition probabilities?

The landmark Aalen-Johansen estimator

For fixed s and ℓ

- ▶ Interested in $P_{\ell m}(s, t) = P(X(t) = m | X(s) = \ell)$
- ▶ Can also be seen as a state occupation probability $P(X(t) = m)$ in a subset
- ▶ Use same subsetting/subsampling idea as used in de Uña Álvarez and Meira-Machado (2015) and Titman (2015)
- ▶ Superscript (LM) denotes counting and at risk processes and estimators based on a landmark data set which selects subjects who are at state ℓ at time s
- ▶ Dependence on s and ℓ suppressed in notation

The landmark Aalen-Johansen estimator

- ▶ Landmark based version of $\hat{\Lambda}(t)$:

$$\hat{\Lambda}_{jk}^{(\text{LM})}(t) = \int_0^t \bar{Y}_j^{(\text{LM})-1}(u) d\bar{N}_{jk}^{(\text{LM})}(u)$$

- ▶ LMAJ estimator given by

$$\hat{P}_{\ell m}^{\text{LMAJ}}(s, t) = \hat{\pi}_{\ell}^{(\text{LM})}(s) \prod_{s < u \leq t} \left(\mathbf{I} + \Delta \hat{\Lambda}^{(\text{LM})}(u) \right),$$

with $\hat{\pi}_{\ell}^{(\text{LM})}(s) = 1$, and 0 otherwise

Comments

$$\hat{P}_{\ell m}^{\text{LMAJ}}(s, t) = \hat{\pi}^{(\text{LM})}(s) \prod_{s < u \leq t} \left(\mathbf{I} + \Delta \hat{\mathbf{h}}^{(\text{LM})}(u) \right)$$

- ▶ It is the Aalen-Johansen estimator of the state occupation probabilities based on the subset of subjects in state ℓ at time s
- ▶ Conditioning is simply accomplished by taking relevant subset
- ▶ This is also at the core of the landmarking method for dynamic prediction
- ▶ Hence the name **landmark Aalen-Johansen**, henceforth LMAJ

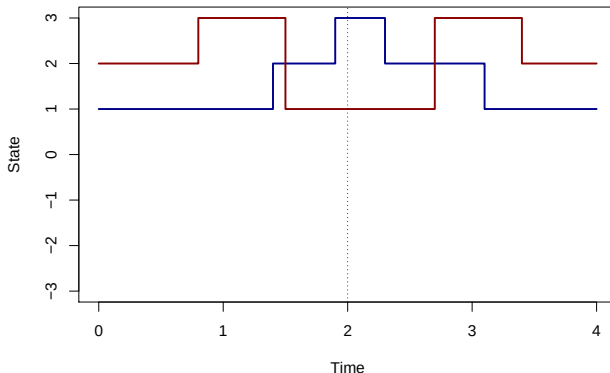
Consistency

Property

- ▶ The LMAJ estimator $\hat{P}_{\ell m}^{\text{LMAJ}}(s, t)$ is a consistent estimator of $P_{\ell m}(s, t)$, provided
 - ▶ The conditions of Datta & Satten (2001) hold for consistency of state occupation probabilities
 - ▶ $P(X(s) = \ell) > 0$

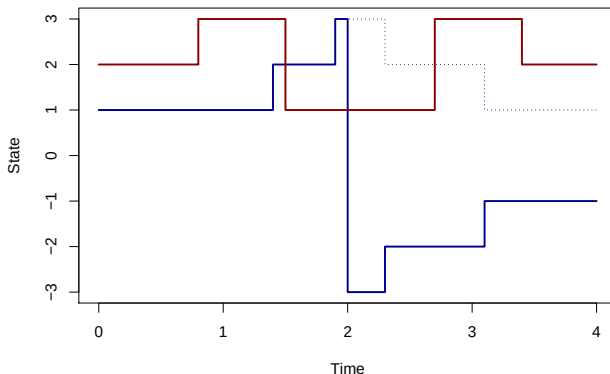
Idea of the proof

- ▶ For instance interested in $P_{\ell m}(s, t)$, for $s = 2, \ell = 1$
- ▶ Define coupled multi-state model $X^*(t)$ derived from original $X(t)$



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Idea of the proof

- ▶ Relation between (transition) probabilities of the two multi-state models

$$P_m^*(t) = P(X^*(t) = m) = P(X(t) = m, X(s) = \ell),$$

- ▶ As a result, the transition probability of interest can be written as

$$P_{\ell m}(s, t) = P(X(t) = m | X(s) = \ell) = \frac{P_m^*(t)}{P_\ell^*(s)}$$

- ▶ A ratio of state occupation probabilities
 - ▶ For which we use the Datta & Satten (2001) result
- ▶ It turns out that the ratio of these state occupation probability estimates is exactly the LMAJ!

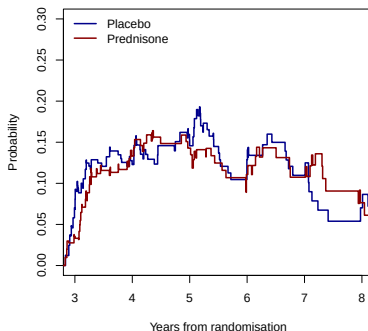
Remarks

- ▶ Standard errors
 - ▶ Greenwood estimators of pointwise standard errors of the Aalen-Johansen estimator of the state occupation probabilities remain valid also if the Markov assumption is violated (Glidden 2002)
 - ▶ Same is true for the LMAJ estimator
 - ▶ For simultaneous confidence bands more elaborate methods need to be used
- ▶ Generalized conditional probabilities
 - ▶ As in Titman (2015), LMAJ can also be generalized to estimates of $P_{\mathcal{L}\mathcal{M}}(s, t) = P(X(t) \in \mathcal{M} \mid X(s) \in \mathcal{L})$
 - ▶ Replace $1\{X_i(s) = \ell\}$ by $1\{X_i(s) \in \mathcal{L}\}$
 - ▶ Sum over all $m \in \mathcal{M}$

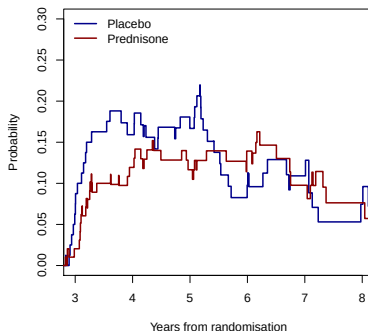
Transition probabilities

Probability of low, given normal at 1000 days

Aalen-Johansen



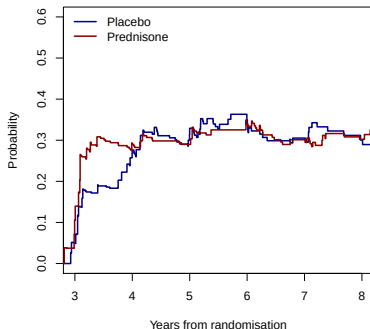
LMAJ



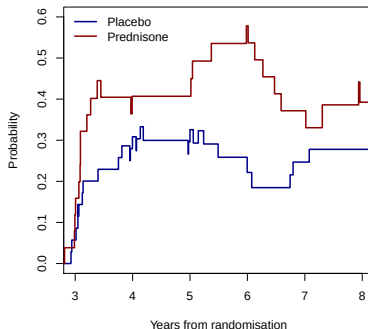
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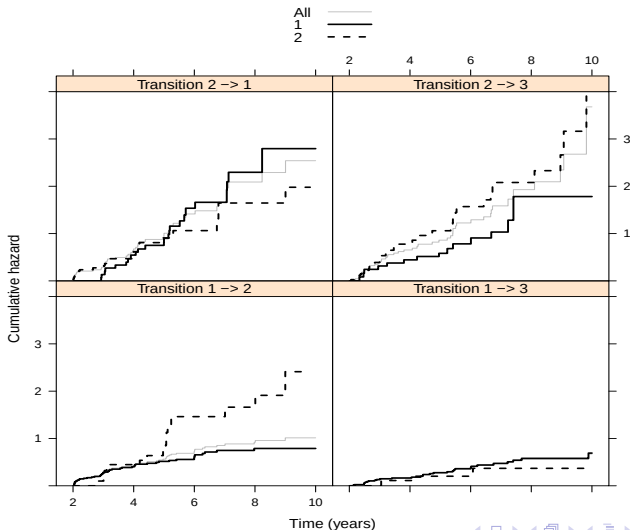


“Half-way” summary

The landmark Aalen-Johansen estimator

- ▶ Simple and intuitive
- ▶ Consistent under same conditions as Datta & Satten (2001)
- ▶ Valid for arbitrary multi-state models
- ▶ Slightly more efficient in simulations than Titman (2015)
- ▶ Probabilities for different target states add up to 1
- ▶ Implemented (function `LMAJ`) in **mstate** package for R (version 0.2.9 and higher)

Estimated cumulative hazards



Estimated cumulative transition hazards

- ▶ Estimated cumulative hazards for all transitions $j \rightarrow k$, separated by state occupied at $s = 2$
 - ▶ For placebo patients
 - ▶ Similar plot for prednisone
- ▶ LMAJ estimator for $P_{1m}(s, t)$ would use only subset 1
- ▶ AJ estimator would use all subjects at risk at time s
- ▶ Under the Markov assumption there would be no difference in hazards after time s between subsets 1 and 2
- ▶ Comparison between hazards can be used for formal test for the Markov assumption

Markov Test

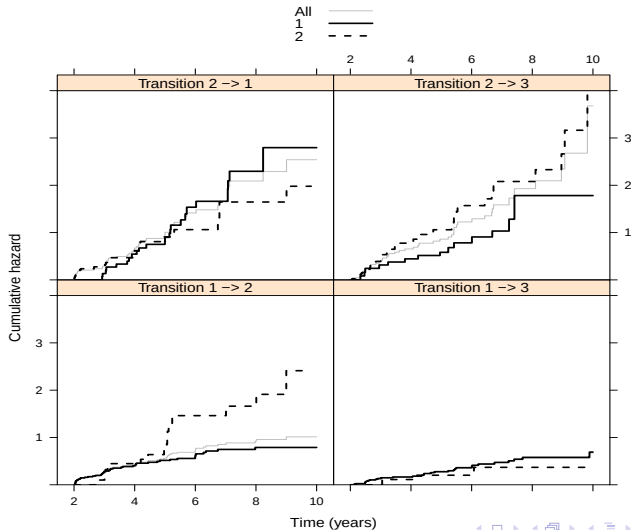
Fix s and ℓ

- ▶ Log-rank test statistic $Z_{jk}^{\ell}(s)$ for each transition $j \rightarrow k$
- ▶ Each log-rank test statistic $Z_{jk}^{\ell}(s)$ has standard normal distribution, if the Markov assumption holds
- ▶ Challenge is to obtain a global test of the Markov assumption
- ▶ Can be achieved by combining the individual log-rank test statistics over transitions $j \rightarrow k$, as well as over qualitative states ℓ and s
- ▶ Simulation studies show good performance
- ▶ Not the focus of this talk

The subset of the LMAJ estimator

- ▶ The LMAJ estimator is robust against violations of the Markov assumption
- ▶ But is based on a subsample of subjects “in the right place at the right time”
- ▶ It is therefore less efficient than AJ, if the Markov assumption is valid
- ▶ Simulation studies show that often, with mild deviations from Markov, AJ has smaller MSE than LMAJ
- ▶ Look again at the estimated transition hazards from $s = 2$ onwards, for subsets 1 and 2
- ▶ Not all transition hazards look different between subsets 1 and 2

Estimated cumulative hazards



Estimated cumulative transition hazards

- ▶ Only for transition $1 \rightarrow 2$ estimated hazards look different between subsets 1 and 2
- ▶ The Markov test mentioned earlier formally confirms this
- ▶ Suggested at workshop: why not use something in between AJ and LMAJ (and maybe get the best of both worlds)?
- ▶ Use all subjects for transitions when we can, and only subsets if needed

Hybrid Aalen-Johansen estimator

Fix state ℓ and time s

- ▶ Define matrix $\hat{\Lambda}^{(H)}$, where $\hat{\Lambda}_{jk}^{(H)}$ equals
 - ▶ $\hat{\Lambda}_{jk}^{LM}$ for transition $j \rightarrow k$ where the Markov test rejects the Markov assumption
 - ▶ $\hat{\Lambda}_{jk}$ for transition $j \rightarrow k$ where the Markov test does not reject the Markov assumption
- ▶ The hybrid Aalen-Johansen estimator is then given by

$$\hat{P}_{\ell m}^{HAJ}(s, t) = \hat{\pi}^{(LM)}(s) \prod_{s < u \leq t} \left(I + \Delta \hat{\Lambda}^{(H)}(u) \right),$$

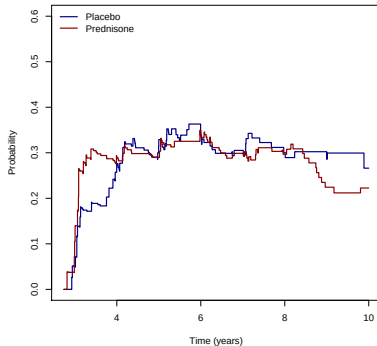
with $\hat{\pi}^{(LM)}(s)$ as for LMAJ

- ▶ When a global test for the Markov assumption is not rejected, one could use the Aalen-Johansen estimator

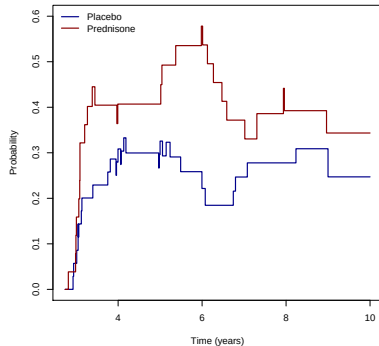
Transition probabilities

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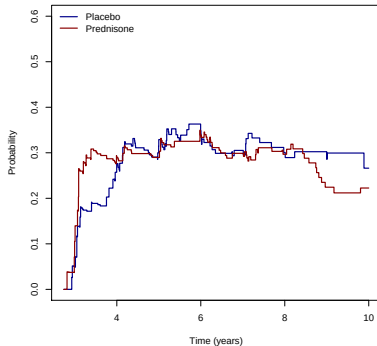
Landmark Aalen-Johansen



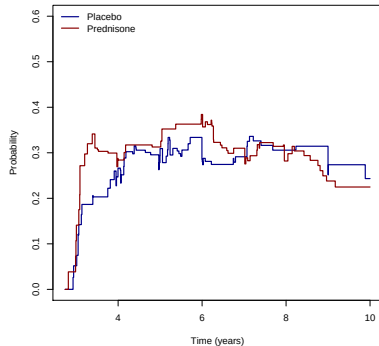
Transition probabilities

Probability of normal, given low at 1000 days

Aalen-Johansen



Hybrid Aalen-Johansen



Discussion

- ▶ Not a lot of work on tests for the Markov assumption
- ▶ Simple test is to include state arrival time (or number of times previous visited) as covariate in Cox model
- ▶ Simulation studies show that it is hard to beat the state arrival time Cox model!
- ▶ Properties of hybrid Aalen-Johansen estimator need to be studied
- ▶ Many extensions of interest
 - ▶ Include covariates
 - ▶ Interval censored (panel) data

References



Aalen OO and Johansen S.

An empirical transition matrix for nonhomogeneous Markov chains based on censored observations.

Scand Stat Theory Appl 1978; 5: 141–150.



Datta S and Satten GA.

Validity of the Aalen–Johansen estimators of stage occupation probabilities and Nelson–Aalen estimators of integrated transition hazards for non-Markov models.

Stat Probab Lett 2001; 55: 403–411.



de Uña Álvarez J and Meira-Machado L.

Nonparametric estimation of transition probabilities in the non-Markov illness-death model: A comparative study.

Biometrics 2015; 71: 364–375.



Titman AC.

Transition probability estimates for non-Markov multi-state models.

Biometrics 2015; 71: 1034–1041.

References



Putter H and Spitoni C.

Non-parametric estimation of transition probabilities in non-Markov multi-state models: The landmark Aalen-Johansen estimator.

Stat Meth Med Res 2018; 27: 2081–2092.



Titman AC and Putter H.

General tests of the Markov property in multi-state models.

Biostat 2022; 23: 380–396.



Maltzahn N, Hoff R, Aalen OO, Mehlum IS, Putter H and Gran JM.

A hybrid landmark Aalen-Johansen estimator for transition probabilities in partially non-Markov multi-state models.

Lifetime Data Anal 2021; 27: 737–760.